AI-AUGMENTED

# Paper Submission & Peer-Review System

*Reimagining a course project with Generative AI*

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# The original project

*A peer-review system I built for a previous web-development course*

### TECH STACK

- Node.js · Express
- EJS server-rendered views
- SQLite (sqlite3 driver)
- bcrypt + express-session
- multer file uploads
- ExcelJS export

### FEATURES

- Author submits manuscript + abstract
- Editor manually assigns reviewers
- Reviewer writes free-text feedback
- Public reader feed of accepted papers

### ROLES & FLOW

1. **AUTHOR**
   Submit manuscript
2. **EDITOR**
   Assign reviewer (random)
3. **REVIEWER**
   Free-text feedback + decision
4. **READER**
   Browse accepted papers (public)

# Limitations of the original

*What an honest audit surfaced — and why GenAI is the right cure*

### Buggy auth wiring

express-session never imported; mixed req.session shape across routes; broken author-id lookup.

### Random reviewer assignment

Math.random() picks any reviewer regardless of expertise — a real workshop never operates this way.

### No integrity checks

No plagiarism detection, no duplicate-submission detection, no AI-generated-text flagging.

### Unstructured reviews

A single textarea per review. No comparable scores, no aggregation, no recommendation field.

### No author tooling

Authors get zero feedback before submitting — no abstract polish, no keyword extraction, no title help.

### No tests, no Docker, no CI

Single-file app.js, hardcoded session secret, two stray .db files. Won't survive code review.

# Why Generative AI?

*Each weakness above maps to something today's AI can directly support*

**4**
AI features planned

**2**
LLM provider backends

**0$**
Zero-cost demo mode

**5**
User roles supported

## Generation

LLMs read an abstract and emit structured prose: review drafts, polished text, alternative titles.

## Retrieval

Embeddings turn text into vectors — reviewer matching and plagiarism become numeric distance problems.

## Decision support

AI never decides. It surfaces signals; humans accept, override, and stay accountable.

# Four planned GenAI features

*Each addresses a specific weakness from the audit*

### Author writing assistant

LLM

Polishes the abstract, suggests alternative titles, extracts keywords on demand. Author edits and accepts.

### Smart reviewer matching

Embeddings

Cosine similarity over reviewer expertise tags. Editor sees a top-3 ranked list with scores and can override.

### Plagiarism + AI-text flag

Embeddings

Max similarity to existing corpus + transparent stylometric AI-text score. Flag, never verdict.

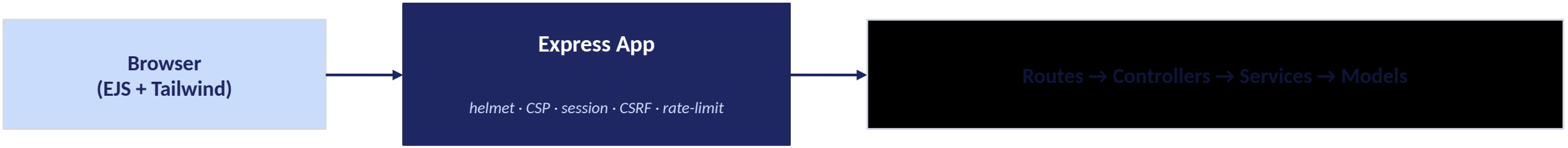
### AI reviewer assistant

LLM

Drafts structured first-pass review (summary / strengths / weaknesses / 1-5 scores / recommendation).

# Architecture I have prepared

*Provider-agnostic LLM abstraction, MVC layout, audit logging*




| reviewerMatcher | plagiarism | aiReviewer | writingAssistant | embeddings |
|---|---|---|---|---|
| *TF-IDF cosine* | *TF-IDF + stylometry* | *LLM draft* | *LLM polish + titles* | *shared vectoriser* |

**LLM PROVIDER SWITCH**

**heuristic (default)**

*Pure JS · offline · zero-cost · ships today*

**claude (when key set)**

*Anthropic Claude Sonnet 4.6 · one-line upgrade*

SQLite (WAL · FK on) · users · papers · reviews · embeddings · ai_audit

*Every AI call logged in ai_audit → ethics auditing & cost tracking*

# Implementation status

*What is scaffolded · what is fully wired · what is intentionally pending*

| Capability | Status | Notes |
|---|---|---|
| **MVC project structure** | DONE | *Routes / controllers / services / models / views all in place* |
| **Auth, sessions, RBAC for 5 roles** | DONE | *helmet, CSP, rate-limit, SQLite session store, role middleware* |
| **Reviewer matching (TF-IDF + cosine)** | DONE | *Working today on the heuristic backend — verifiable in the editor dashboard* |
| **Plagiarism similarity scoring** | DONE | *Running on every submission against the existing corpus* |
| **AI-text stylometric flag** | DONE | *Transparent sub-signals exposed to the editor* |
| **AI reviewer draft (heuristic templates)** | SCAFFOLDED | *UI + API in place; switches to real LLM once API key supplied* |
| **Author writing assistant (heuristic)** | SCAFFOLDED | *Polish / titles / keywords scaffolded; LLM call ready to flip on* |
| **Claude API integration** | PLANNED | *Adapter committed; activates with one env var; deferred until budget approved* |
| **Sentence-transformer embeddings** | PLANNED | *Roadmap item for Assessment 2 — replaces TF-IDF for richer matching* |
| **User study + benchmarking** | PLANNED | *Planned for Assessment 2; ai_audit table already collects the data* |

# AI tools I used to build this

*How I shipped a 36-file codebase, a report, and this deck in days*

### Claude

*Sonnet 4.6 (chat + code)*

Architecture decisions, full code generation, the 1500-word report, this presentation. The main brain throughout.

**main coding partner**

### Whisper Flow

*voice → text dictation*

Talked through prompts and design ideas instead of typing. Way faster turnaround when iterating on architecture.

**prompt at speaking speed**

### Cursor

*AI-native code editor*

Inline AI completion and refactoring while editing the EJS views and controller code. Tab-to-accept, repeat.

**inline edits**

*Also in the chain · ChatGPT for second opinions · GitHub Copilot for repetitive code · Excalidraw for diagrams · Pandoc for docx checks*

# Ethical considerations

*Peer review affects careers — we engineer for human accountability*

### Authorship & accountability

AI drafts; humans submit. ai_assisted flag on every review. The model never autonomously approves anything.

### Bias & fairness

LLM bias inherited from training data; expertise tags can under-represent niche fields. Editor override + audit trail mitigate.

### Privacy & data minimisation

Defaults to local heuristic backend. When Claude is on, only title + abstract leave the trust boundary. Every external call audited.

### AI-text detection limits

Stylometric score is a flag, not a verdict. Sub-signals are exposed; submission is never blocked solely on this score.

# Live demo: the recreated v2

*Already running — five role-aware dashboards with AI affordances*

**1** **Author submit page**

*Three AI buttons (Polish · Suggest titles · Extract keywords) sit alongside the abstract.*

**2** **Editor dashboard**

*Top-3 AI-ranked reviewers per paper with cosine score, plus manual override and final decision.*

**3** **Reviewer review**

*"Generate AI draft" populates summary / strengths / weaknesses / scores / recommendation.*

**4** **Public reader feed**

*Newspaper-style grid of accepted articles, full-text search, public download — no login.*

TO RUN LOCALLY

```
$ cd paper-submission-system-v2
$ npm install
$ npm run setup
$ npm start

→ http://localhost:3000

Demo accounts (Password123!):
 alice / bob — author
 reviewer_ml / db / se
 editor — manual + AI assignment
 admin — audit dashboard
 reader — public feed
```

*Default mode: heuristic backend (zero cost). Add ANTHROPIC_API_KEY + LLM_PROVIDER=claude to upgrade — no other code changes.*

# Roadmap to Assessment 2

*A five-step plan from today through the Assessment 2 submission*


### Now

v2 codebase live · heuristic backend covers all four AI features end-to-end · audit table collecting usage

### Week 1-2

Activate Claude provider · benchmark draft quality vs heuristic on 10 sample papers · cost & latency telemetry

### Week 3-4

Replace TF-IDF with sentence-transformers (@xenova/transformers, local) · re-evaluate matching precision

### Week 5-6

Add conflict-of-interest detector (institutional + co-authorship) · automated tests for regression

### Submission

Small user study with peers comparing reviewer thoroughness with vs without AI draft assistance

*Takeaway: every Assessment-2 experiment has scaffolding waiting for it — the v2 codebase is the platform, not the prototype.*

THANK YOU

# Questions, suggestions, or budget approvals welcome.

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*Repo: paper-submission-system-v2 · Report: Assessment_Report.docx*